

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Dept: CSE - AIDS | Course: DSA06 - Data Handling and Visualization

Set 3: Employee Performance Evaluation

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Dataset: Employee ID, Department, Years of Service, Performance Score for 5 employees.

Question 1

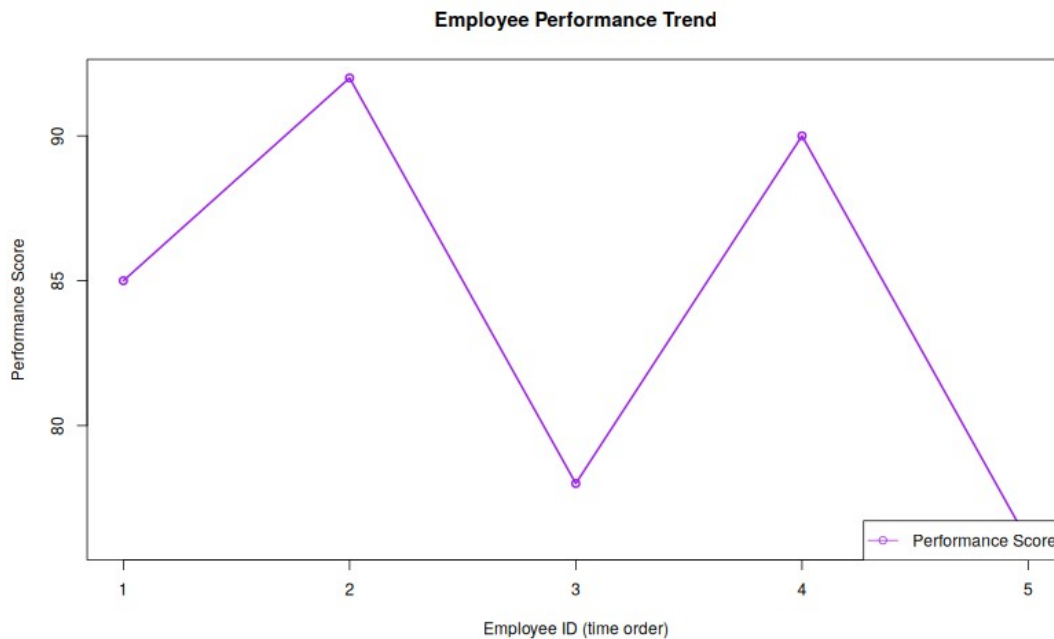
Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Code

```
emp <- data.frame(
  EmployeeID = 1:5,
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsOfService = c(5, 3, 7, 4, 2),
  PerformanceScore = c(85, 92, 78, 90, 76)
)

plot(emp$EmployeeID, emp$PerformanceScore, type="o", col="purple", lwd=2,
      xlab="Employee ID (time order)", ylab="Performance Score",
      main="Employee Performance Trend")
legend("bottomright", legend="Performance Score", col="purple", lty=1, pch=1)
```

Output



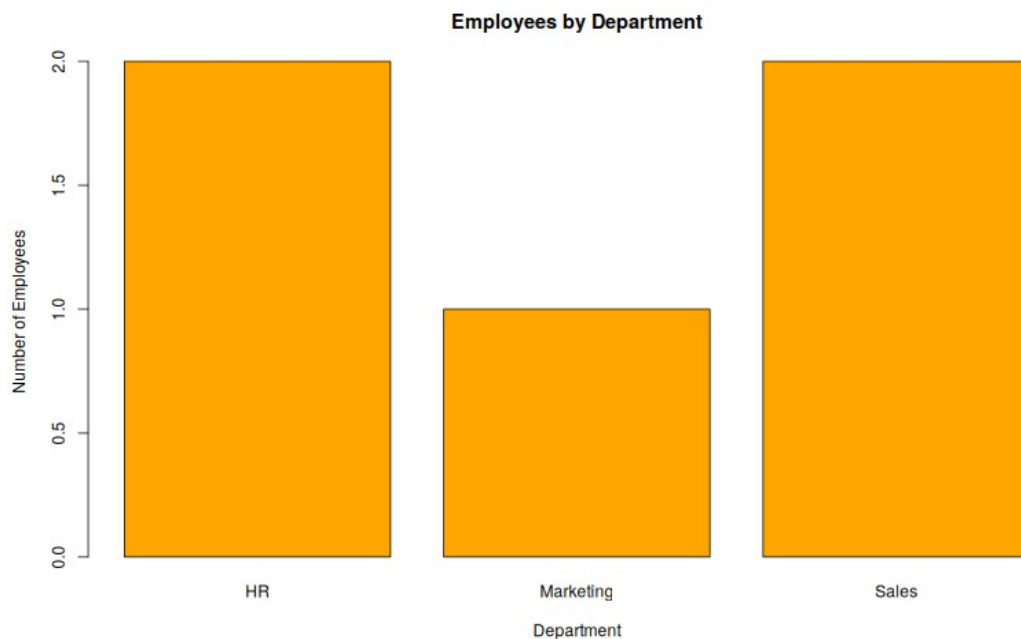
Question 2

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Code

```
barplot(table(emp$Department), col="orange",  
        xlab="Department", ylab="Number of Employees",  
        main="Employees by Department")
```

Output



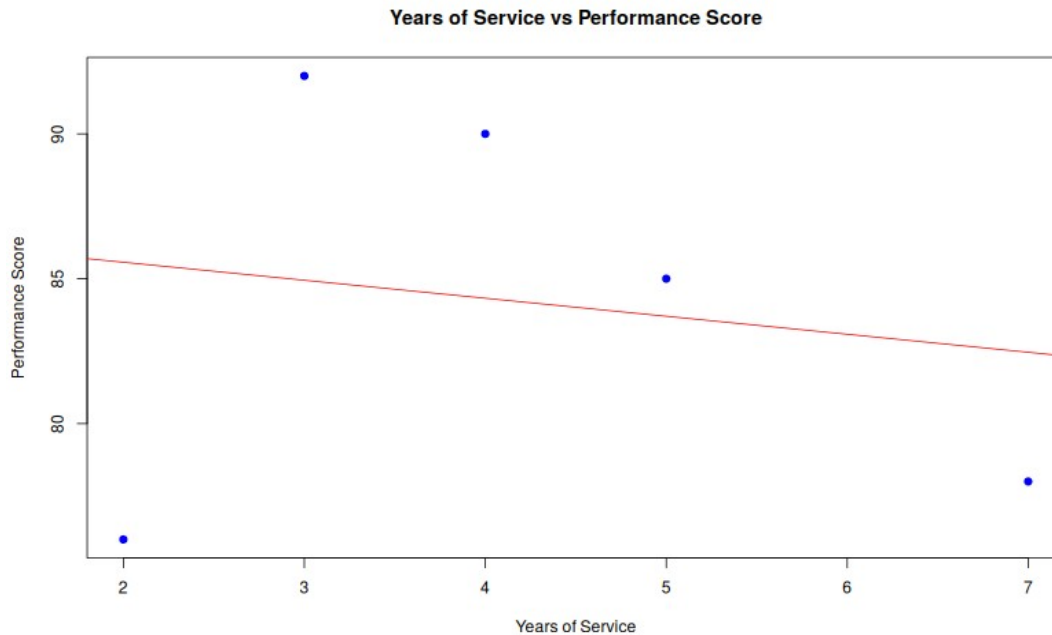
Question 3

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

Code

```
plot(emp$YearsOfService, emp$PerformanceScore, pch=19, col="blue",  
     xlab="Years of Service", ylab="Performance Score",  
     main="Years of Service vs Performance Score")  
abline(lm(PerformanceScore ~ YearsOfService, data=emp), col="red")
```

Output



Insight: The points don't follow a strong single trend (HR employees with fewer years score high, Marketing with the most years scores lowest), suggesting performance depends more on department/role than years of service alone.

Question 4

Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Code

```
-- Tableau steps (build this in Tableau Desktop) --  
1. Import the Employee Performance data into Tableau.  
2. Sheet 1: Employee ID (Columns) vs Performance Score (Rows), Line mark.  
3. Sheet 2: Department (Columns) vs COUNT(Employee ID) (Rows), Bar mark.  
4. Create a new Dashboard, drag both sheets onto it.  
5. Add a Department filter and apply it to both sheets for interactivity.
```

Output



Note: Tableau Desktop isn't available in this environment, so the panel above was rendered in R to preview the same layout (line chart + bar chart together) that the Tableau dashboard would display once built with the steps above.